

Docker Permission Denied Error SOP

Objective

Resolve Docker permission errors that prevent users, applications, or containers from running Docker commands or accessing mounted files.

Scope

Applicable to Docker-based services, containerized application deployments, DevOps teams, SRE teams, platform teams, and synthetic incident workflows across development, staging, and production-like environments.

Pre-requisites

- 1. Access to Docker host, Docker CLI, container runtime logs, and the target container or image name.
- 2. Permission to inspect containers, images, volumes, networks, resource limits, and compose files where applicable.
- 3. Access to application logs, deployment artifacts, Dockerfile, environment variables, and registry credentials.
- 4. Monitoring access for host CPU, memory, disk, container restart count, and container health checks.
- 5. Escalation contact for application, DevOps, security, registry, infrastructure, and platform teams.

Incident Metadata

Field	Value
Ticket ID	T016
Issue	Docker permission denied error
Category	Security
Service	Docker
Severity	Medium
Status	Resolved
Description	User cannot run docker commands
Expected Resolution	Add user to docker group

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L014	5/1/2026 10:07	ERROR	Permission denied error	DOCKER_PERM
L034	5/1/2026 10:17	ERROR	Permission denied error	DOCKER_PERM

Procedure

- 1. Capture the exact permission denied message, user, host, command, container, and file path involved.
- 2. Check whether the user has permission to access the Docker socket or run Docker CLI commands.
- 3. Validate user membership in the docker group or approved sudo access model.
- 4. Inspect file and directory permissions for bind mounts and container-mounted paths.
- 5. Check container user, UID, GID, and security options in Dockerfile or compose file.
- 6. Apply the approved permission fix such as group membership, sudo policy, file ownership, or container user mapping.
- 7. Log out and log back in if group membership was changed.
- 8. Rerun the failing command and confirm access is restored.

Common Commands

```
docker ps -a
docker logs <container-name-or-id> --tail=100
docker inspect <container-name-or-id>
docker stats --no-stream
docker images
docker network ls && docker network inspect <network-name>
docker volume ls && docker volume inspect <volume-name>
docker system df
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
User cannot run docker	User not in docker group	Add user to docker group after approval
Bind mount denied	Host file permissions	Fix ownership or ACL
Container cannot write file	Container user mismatch	Align UID/GID or mount permissions
Docker socket denied	Socket permission issue	Use approved access method

Best Practices

1. Do not grant Docker access without security approval.
2. Prefer least-privilege access patterns.
3. Use explicit container users for production images.
4. Document ownership requirements for bind mounts.

References

Docker Docs | Internal Tickets CSV | Internal Logs CSV | Container Runtime Logs | Monitoring Dashboards | DevOps Runbook